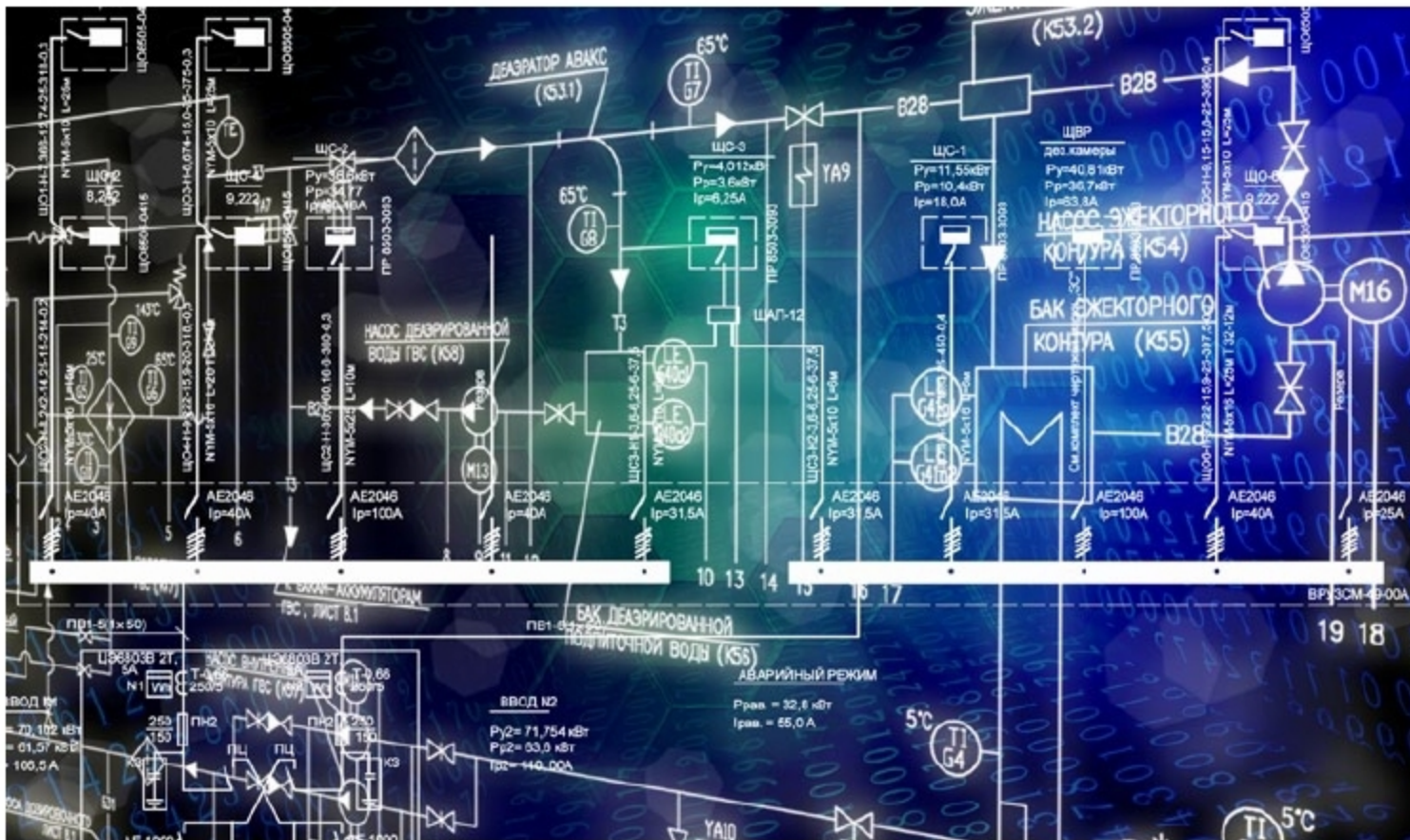




Behind Every Successful Product There Is A Good Design And Engineering

The author summarises a set of issues that need to be addressed if a product has to succeed in the market in its initial release itself



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In my career I have developed many products, ranging from telecom to avionics, and all of them have been manufactured successfully. However, in the last few years of my interactions with startups and SMEs, I have observed that none of them really know what it takes to manufacture and, sometimes, design a reliable product. Having been involved in helping a dozen startups involved in hardware design and manufacturing, I have seen most of them fail when they go to the market first—most of them come to me after they are hit by the first failure. One common problem I have seen is that they rushed to the market because the first

customer wanted a trial, or because of the first revenue opportunity, but the design was not ready for real-life deployment.

The product gets into all kinds of problems once deployed and the customer gets disillusioned due to those issues. One of the problems I have noticed is converging to a lack of product engineering. When the startup takes its product to an electronics manufacturing services (EMS) vendor—most of the times these vendors are an assembly house with a purchase department called supply chain—product engineering is completely missing. What I mean by engineering is designing the PCBs with proper DfX inputs, and having a manufacturing tester